

Question number	Answer	Additional guidance	Mark
4(a)(i)	1.27 / 1.33 (1)	DO NOT ACCEPT 1.33 recurring	(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	21 / 21.3 / 21.26 / 24.8 / 24.81 / 25 (%) (1)	Allow ecf from (i)	(1)

Question number	Answer	Mark
4(b)(i)	The only correct answer is B. <i>A is incorrect because S is the oldest ring</i> <i>C is incorrect because P is the newest ring and S is the oldest ring</i> <i>D is incorrect because P is the newest ring and S is the oldest ring</i>	(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes three of the following points: • because each year a ring will be formed (1) • and the thickness of the rings will {be different for each year / depend on the conditions} (1) • the rings in each sample can be {matched / lined up} (1) • and only counting a ring once if it overlaps (1)	ACCEPT the number of rings is the age this tree is {68 to 76} years old quoted figures e.g. 69 rings = 69 years ACCEPT named condition e.g. temperature	(3)

Question number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes the following points: • measure the height (of the whole tree) (1) • height divided by the {(total) number of rings / age} (1)	ACCEPT length for height measure {radius / diameter / total width of rings} at bottom (of tree) ACCEPT {radius / diameter} at bottom of tree divided by {number of rings / age} CE from mp 1	(2)